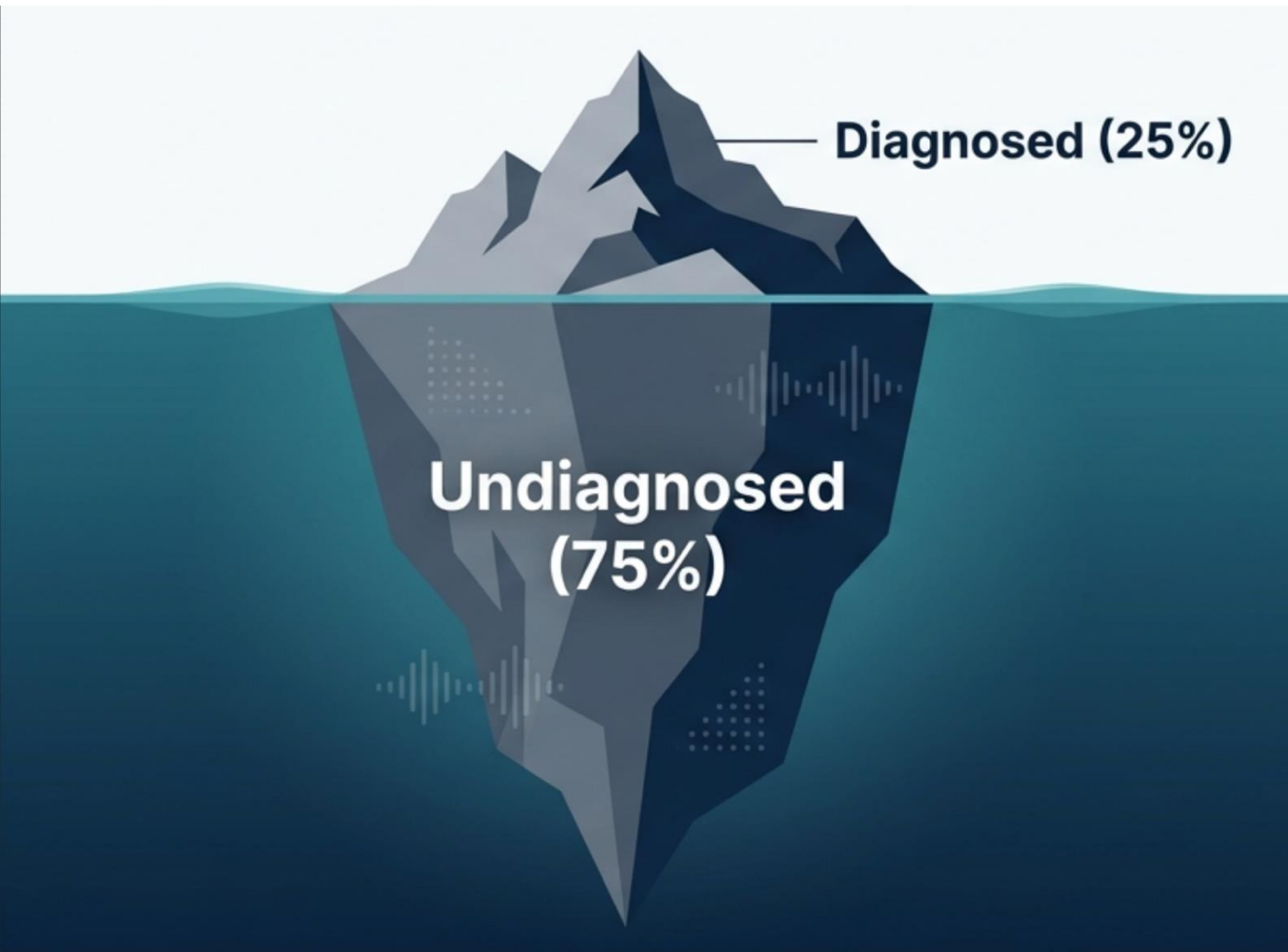


Attention Feature Fusion with Cluster Contrastive Learning for Snoring and Breath-Holding Detection Using Seismic Sensing

Yingjian Song*, Jiayu Chen*, Zixuan Zeng, Yida Zhang,
Zaid Farooq Pitafi, Bradley G. Phillips, Xiang Zhang,
Fei Dou, Wenzhan Song

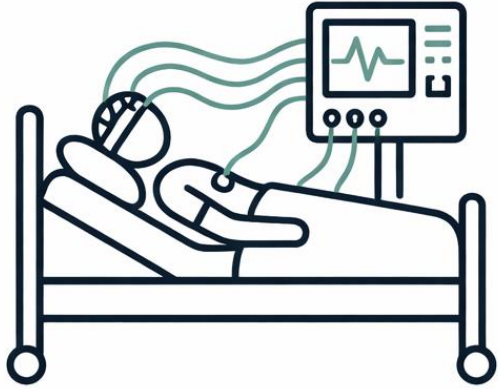
Background



Snoring and breath-stopping essentially reflect the degree of airflow limitation or obstruction in the upper airway.

These events directly reflect the *stability of the respiratory system during sleep*. They are not only key indicators for diagnosing sleep apnea, but are also associated with cardiovascular diseases, metabolic disorders, as well as daytime sleepiness and cognitive decline.

Background



Polysomnography

Gold Standard

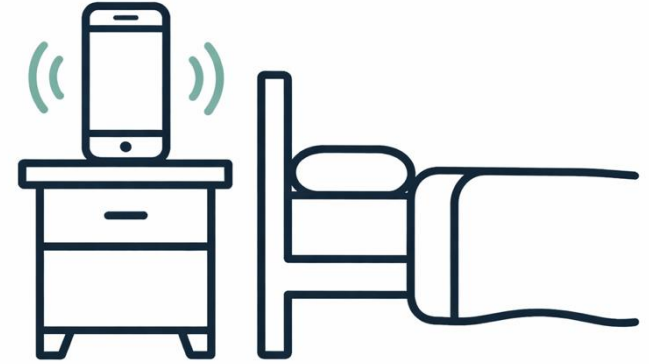
Accurate but requires a lab setting and wired sensors. Expensive and inaccessible.



Wearables

Intrusive

Require continuous body contact causing discomfort. Dependent on battery life.



Cameras & Audio

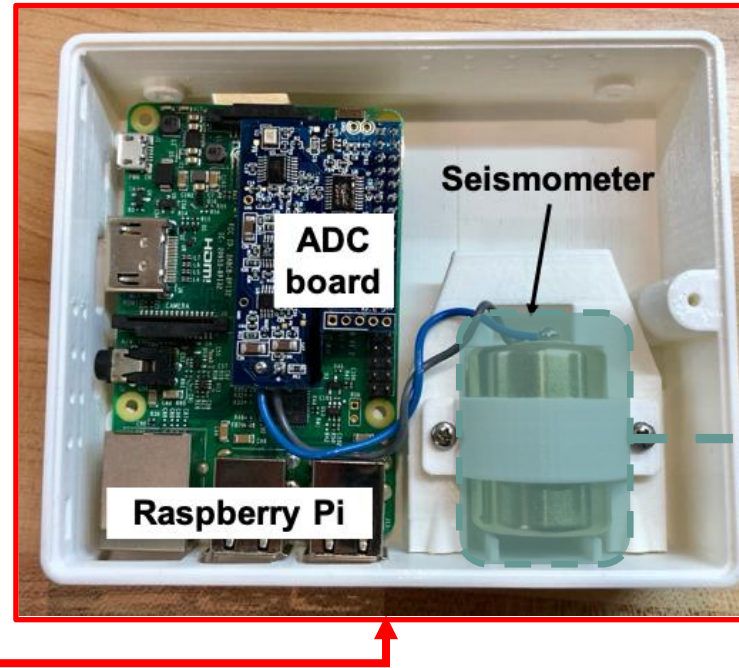
Privacy Concerns

Sensitive to positioning, lighting conditions, and blankets.

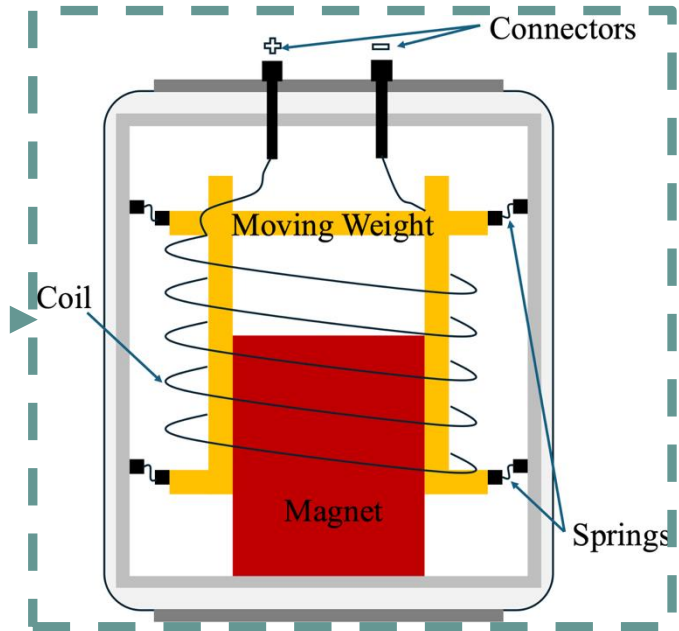
Sensor Installation



Under Bed Deployment



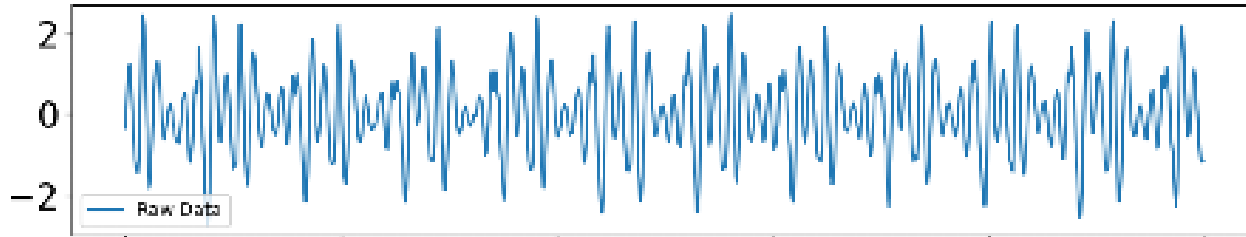
BedDot



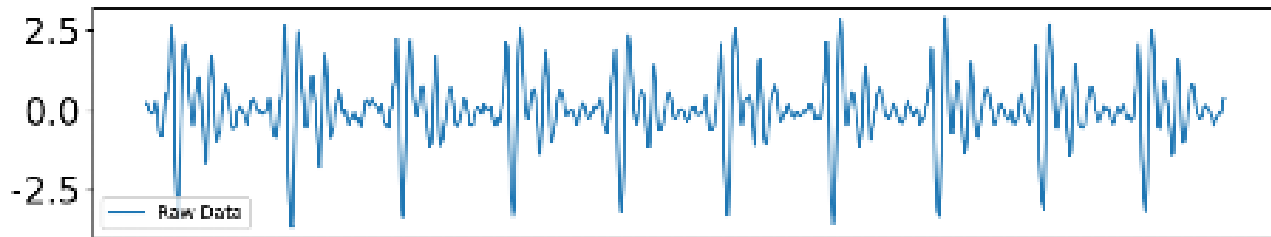
Geophone

Signal Analysis

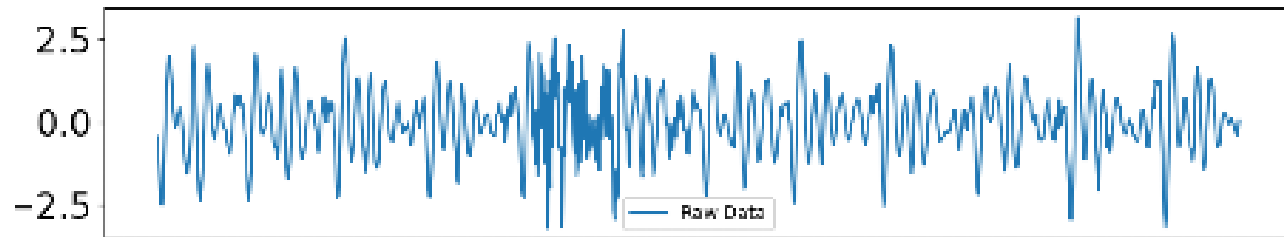
Normal Breath



Breath Stop



Snore Light



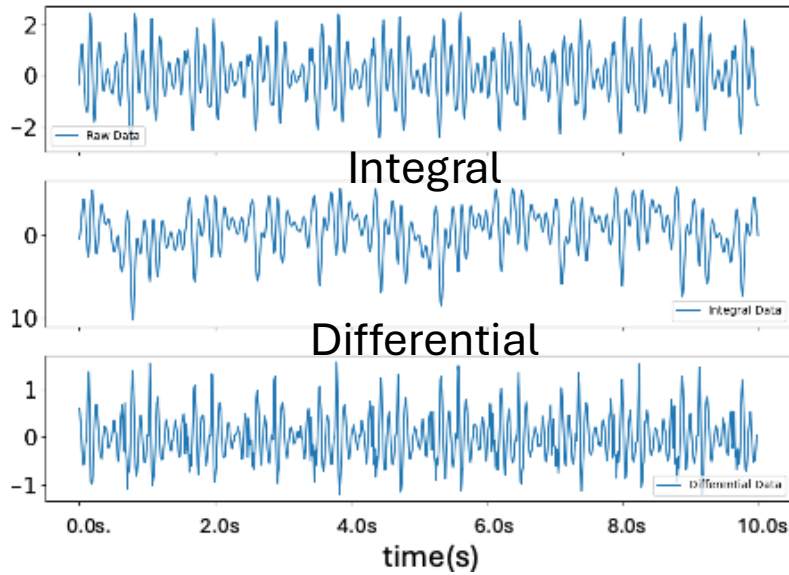
Bodyseismography(BSG) is complex mixture of heartbeats, respiration, and movements.

Breath-Stopping Signals: Clear heartbeat patterns with little or no respiratory modulation.

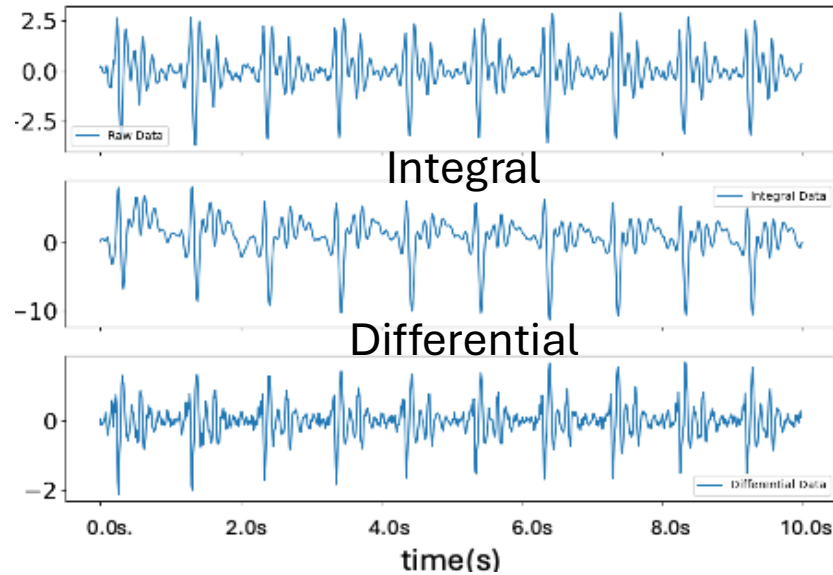
Snoring Signals: Characterized by prominent high-frequency vibrations.

Signal Pre-Processing

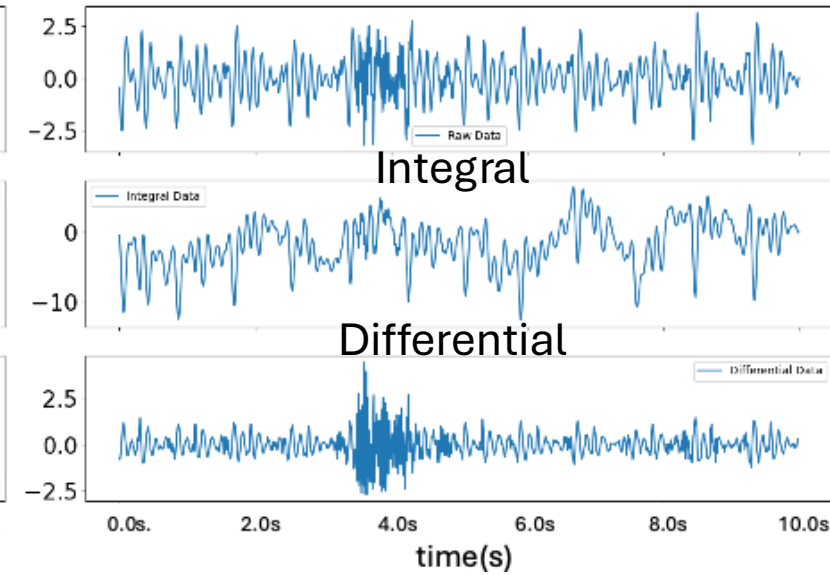
Normal Breath



Breath Stop



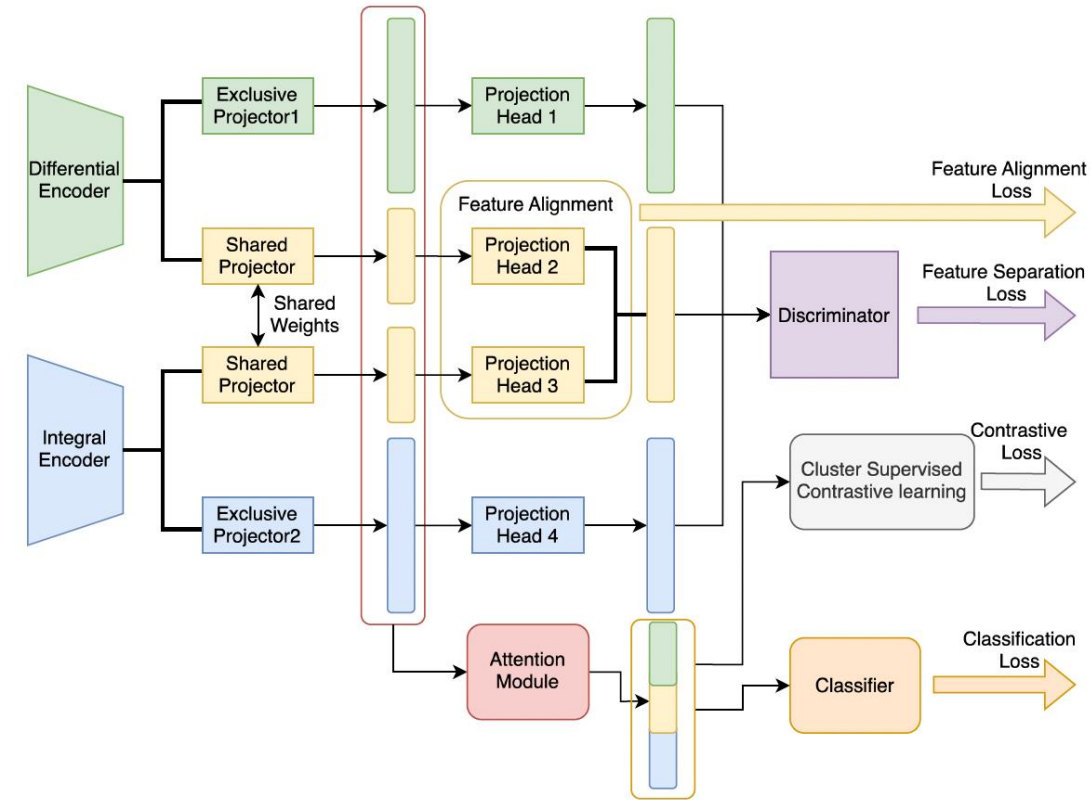
Snore Light



Utilize the Integration to enhance the respiration waveform.

Utilize the Differentiation to enhance the high-frequency snoring features.

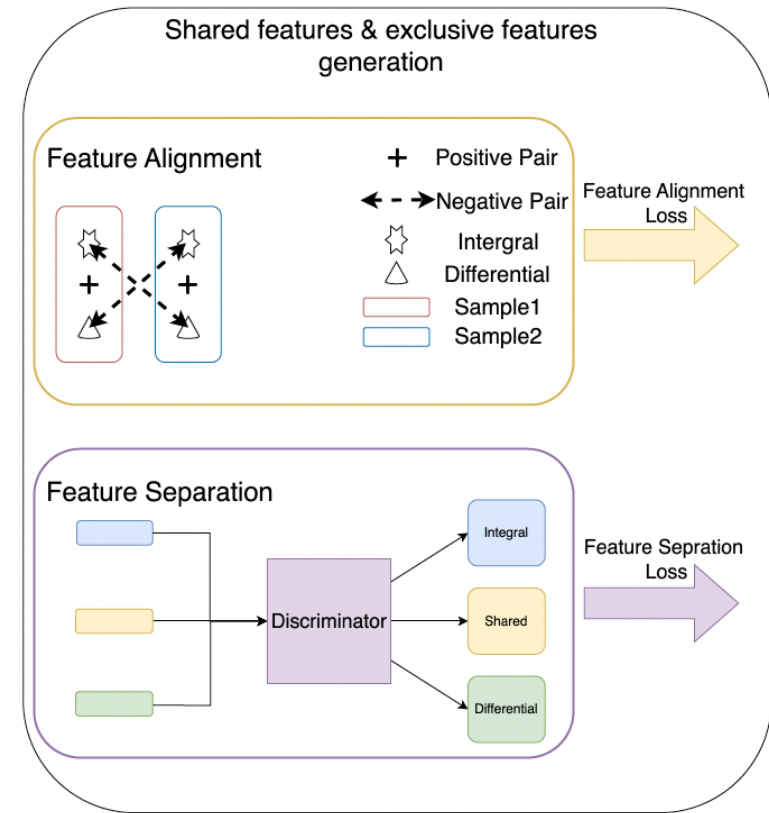
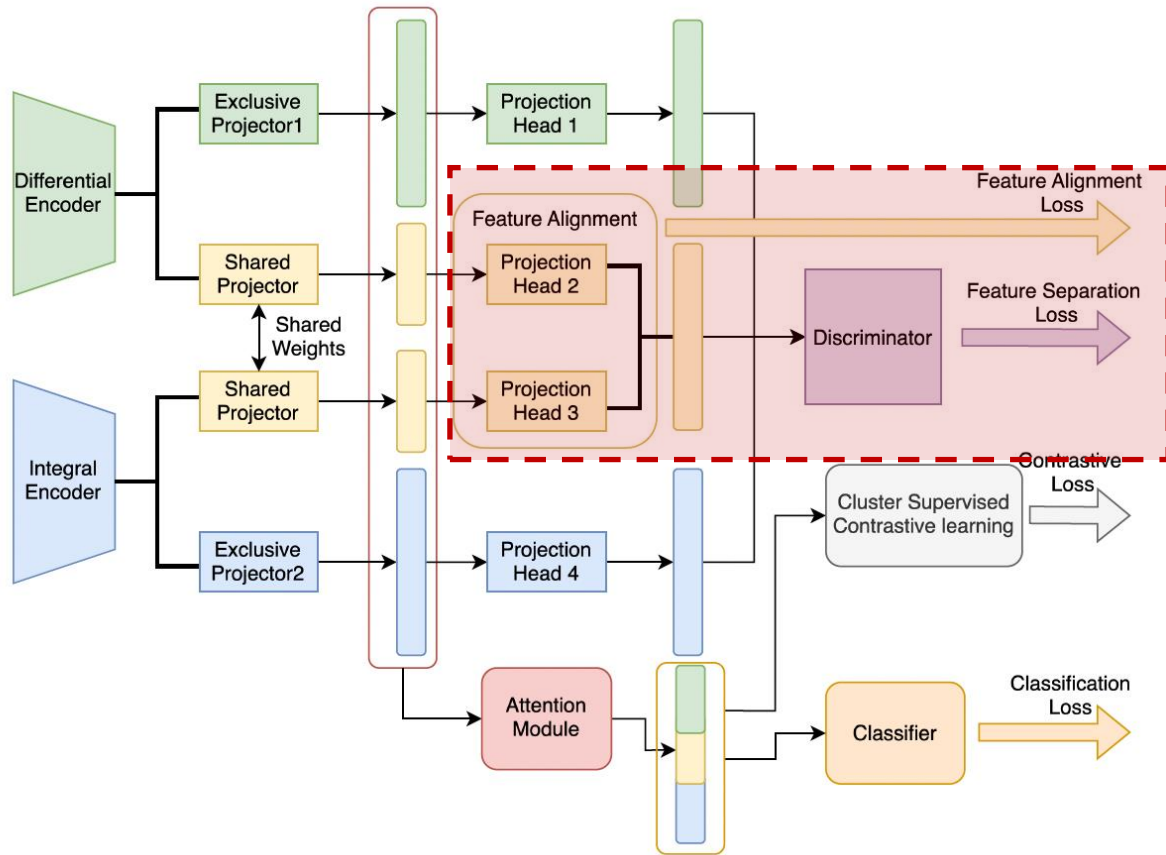
Framework Overview: AFFCL



The integral/differential signals are fed into the Attention Feature Fusion and Cluster Contrastive Learning (AFFCL) framework, which consists of:

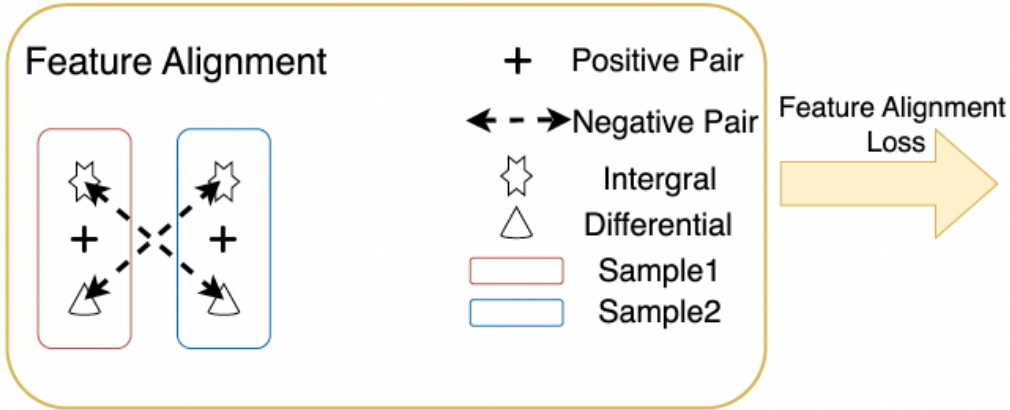
1. Shared Features and Exclusive Features Generation
2. Attention Score Guidance
3. Cluster Supervised Contrastive Learning

AFFCL: Shared Features and Exclusive Features Generation



This module projects integral and differential signals into **shared** and **exclusive** spaces. It consists of two components: **Feature Alignment** and **Feature Separation**.

AFFCL: Shared Features and Exclusive Features Generation



Embeddings from the same segment should be close (positive pairs), while embeddings from different segments or views should be far apart (negative pairs) in the embedding space.

In the Feature Alignment module, contrastive learning is used to align integral and differential signals in a shared embedding space.

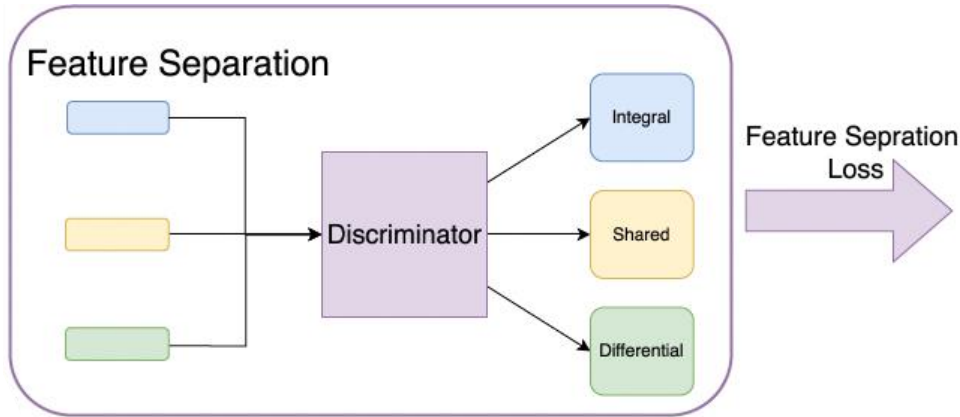
Specifically, InfoNCE loss is used to pull positive pairs closer and push negative pairs apart.

$$\ell_i^{IS} = -\log \frac{\exp(\frac{\text{sim}(Z_i^{IS}, Z_i^{DS})}{\tau})}{\sum_{j=1}^N \mathbb{1}_{i \neq j} \exp(\frac{\text{sim}(Z_i^{IS}, Z_j^{DS})}{\tau})}$$

$$\ell_i^{DS} = -\log \frac{\exp(\frac{\text{sim}(Z_i^{DS}, Z_i^{IS})}{\tau})}{\sum_{j=1}^N \mathbb{1}_{i \neq j} \exp(\frac{\text{sim}(Z_i^{DS}, Z_j^{IS})}{\tau})}$$

$$\mathcal{L}_{FA} = \frac{1}{2N} \sum_{i=1}^N (\ell_i^{IS} + \ell_i^{DS})$$

AFFCL: Shared Features and Exclusive Features Generation



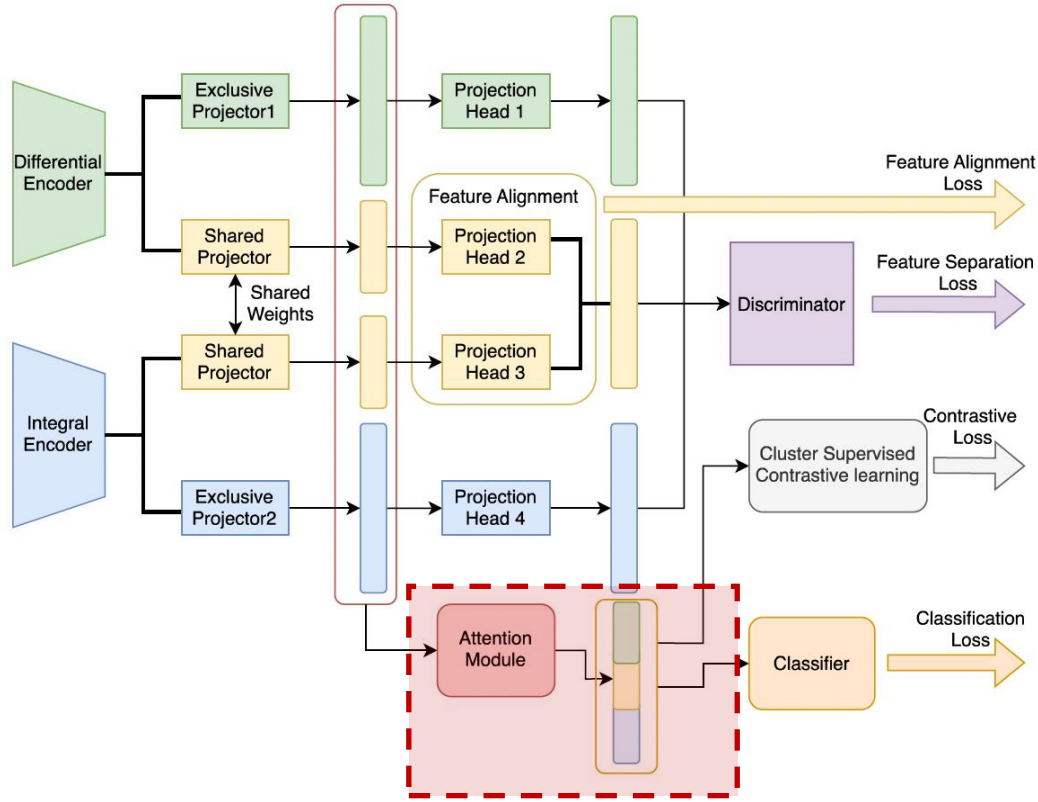
Moreover, different views of the same segment (integral-exclusive, differential-exclusive, and shared) contain distinct information.

The Feature Separation module maximizes the separation between exclusive and shared features by applying a discriminator D . Specifically, this is implemented as a three-class classification task:

$$\mathcal{L}_{FS} = \frac{1}{3N} \sum_{k=1}^{3N} \sum_{c=1}^3 y_{k,c} \log(\hat{y}_{k,c})$$

$y_{k,c}$ is the ground-truth one-hot label, indicating which feature space the k -th embedding belongs to.

AFFCL: Attention Score Guidance



Embeddings from different spaces may carry different information. Self-attention is utilized to combine these embeddings with different weights.

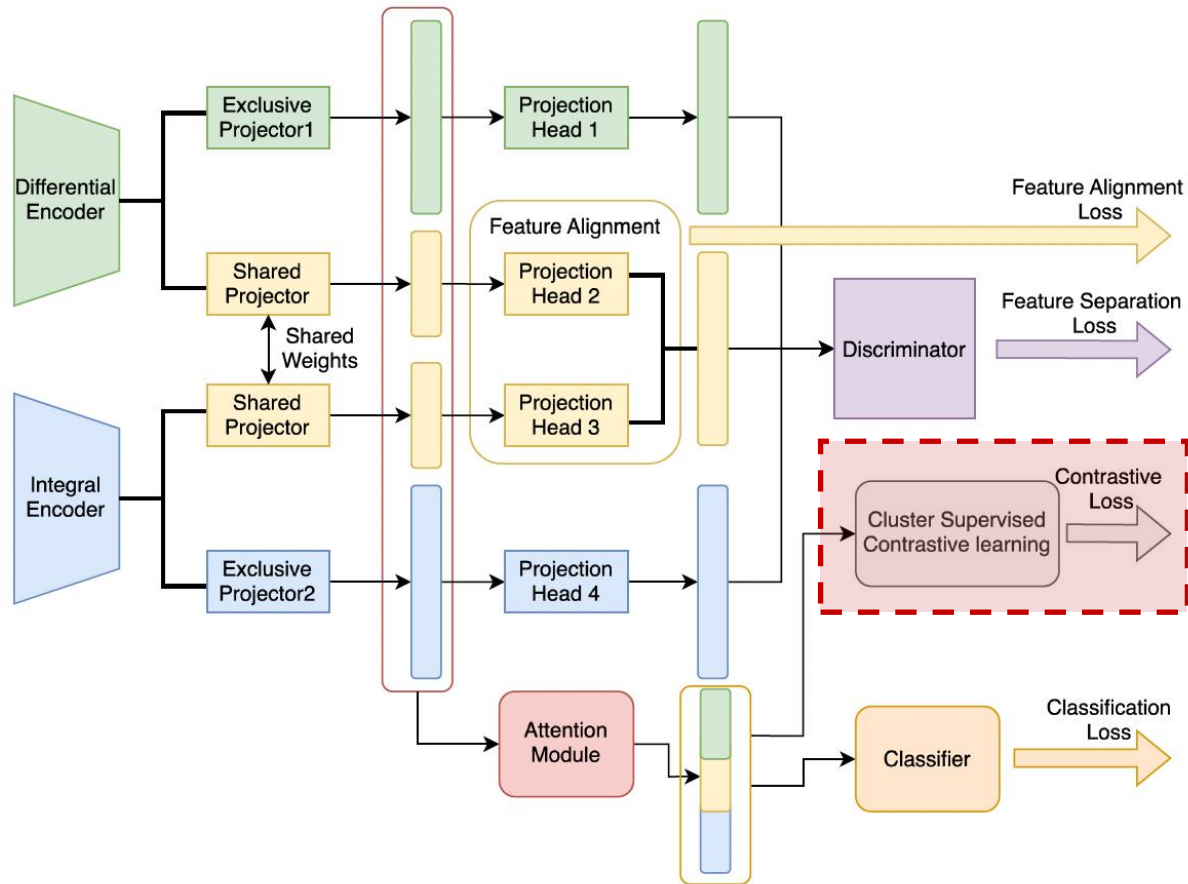
The guided attention score loss serves as a regularization term encouraging segments from the same class to exhibit similar attention distributions.

$$\mathcal{L}_{score} = \frac{1}{|C|} \sum_{c \in C} \frac{1}{|I_c|} \sum_{i \in I_c} \|M_i - M_c\|$$

$$M_i = [score_I^i, score_D^i, score_S^i] \quad M_c = \frac{1}{|I_c|} \sum_{i \in I_c} M_i$$

M_i denotes the attention score vector for segment i , while M_c represents the centroid of the attention scores for class c .

AFFCL: Cluster Supervised Contrastive Learning



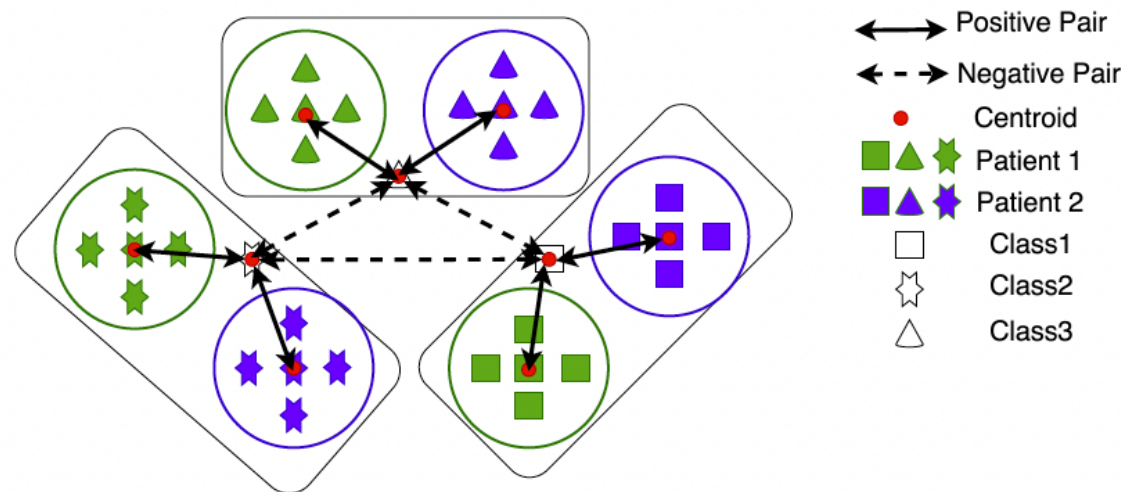
How to deal with real-world patient variability?

Supervised contrastive learning (SCL) optimizes the model at the segment level.

In contrast, Cluster-Supervised Contrastive Learning (CSCL) operates at a higher, cluster level.

AFFCL: Cluster Supervised Contrastive Learning

Cluster supervised contrastive learning



Specifically, CSCL treats samples from the same patient and the same class as a local cluster and computes a centroid for each cluster.

Clusters belonging to the same class but originating from different patients are then consolidated by drawing their centroids, whereas centroids of clusters from different classes are pushed apart.

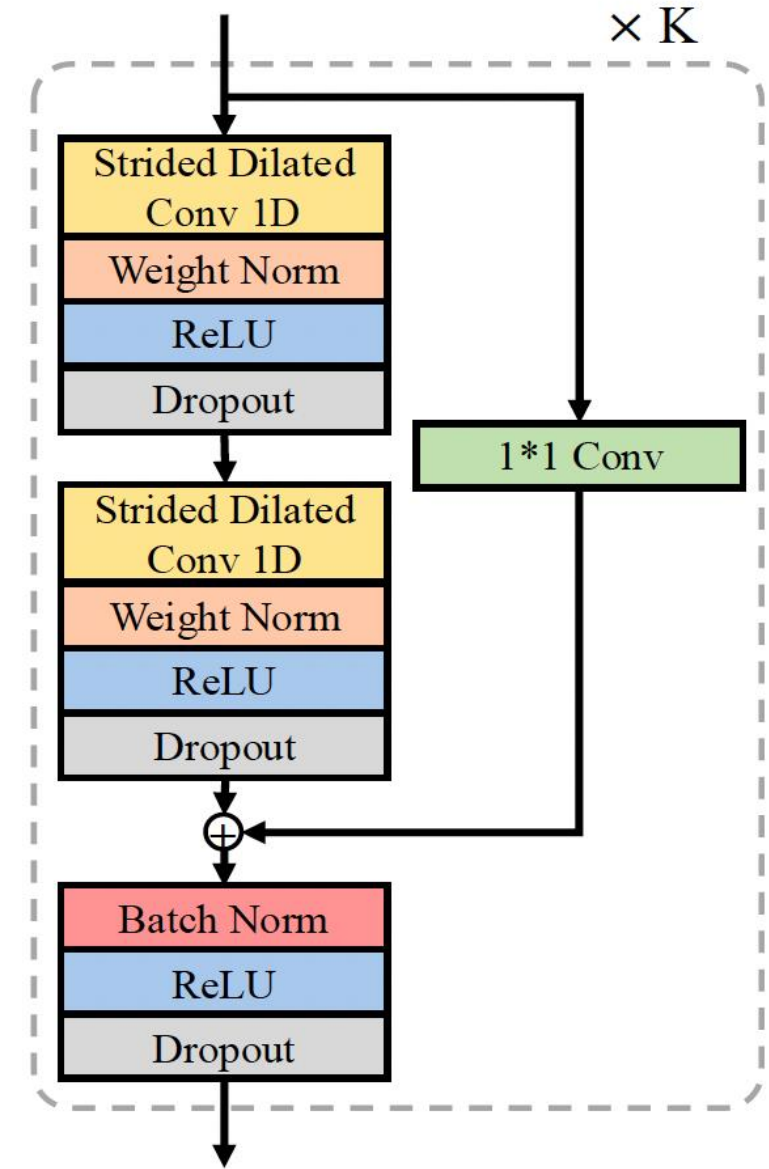
CSCL maximizes the similarity of embeddings obtained from different patients within the same class, while maximizing the separation between different classes regardless of patient identity.

$$\mathcal{L}_C = -\frac{1}{|C|} \sum_{c \in C} \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \log \frac{\exp(\bar{z}_c^T z_c^p / \tau)}{\sum_{c' \in C \setminus \{c\}} \exp(\bar{z}_c^T \bar{z}_{c'})}$$

AFFCL: Training

The final training loss consists of: Feature alignment loss $\mathcal{L}_{FA}$, feature separation loss $\mathcal{L}_{FS}$, a guided attention score loss $\mathcal{L}_{score}$, cluster supervised contrastive loss $\mathcal{L}_C$, breathing status classification loss $\mathcal{L}_{task}$.

$$\mathcal{L} = \lambda_1 \mathcal{L}_{FA} + \lambda_2 \mathcal{L}_{FS} + \lambda_3 \mathcal{L}_{score} + \lambda_4 \mathcal{L}_C + \lambda_5 \mathcal{L}_{task}$$



Data Collection

Lie and rest (1 min)

Snoring simulation:

High intensity (20 s) → rest (15–30 s)

Low intensity (20 s) → rest (15 s)

Breath-holding:

Deep breath hold (≤ 20 s) → rest (15 s)

Normal breathing → breath hold (5–10 s) → rest (5 min)

Switch to foam bed and repeat the same protocol

Hospital & Foam bed

50 subjects (32M/18F; 6 excluded)

10-second windows

Normal: 12,105

Breath-stopping: 1,268

Snoring: 4,188

Cross subject five-fold validation

Overall Performance

Table: Overall Performance

Class	Metric	Fold1	Fold2	Fold3	Fold4	Fold5	Ave
ALL	Acc	93.85	93.12	91.04	93.26	95.72	93.40
	F1	90.70	93.20	93.49	91.64	93.09	92.42
BRS	Acc	96.51	86.67	78.72	88.65	97.13	89.54
NBR	Acc	91.67	97.08	99.30	95.20	97.03	96.06
SNR	Acc	93.36	95.62	95.09	95.93	93.01	94.60

These results indicate that the overall performance of AFFCL shows strong performance for breath status classification and great potential for assisting in the identification of sleep apnea.

All: performance across all three classes

BRS: breath-stopping

NBR: normal breathing

SNR: snoring

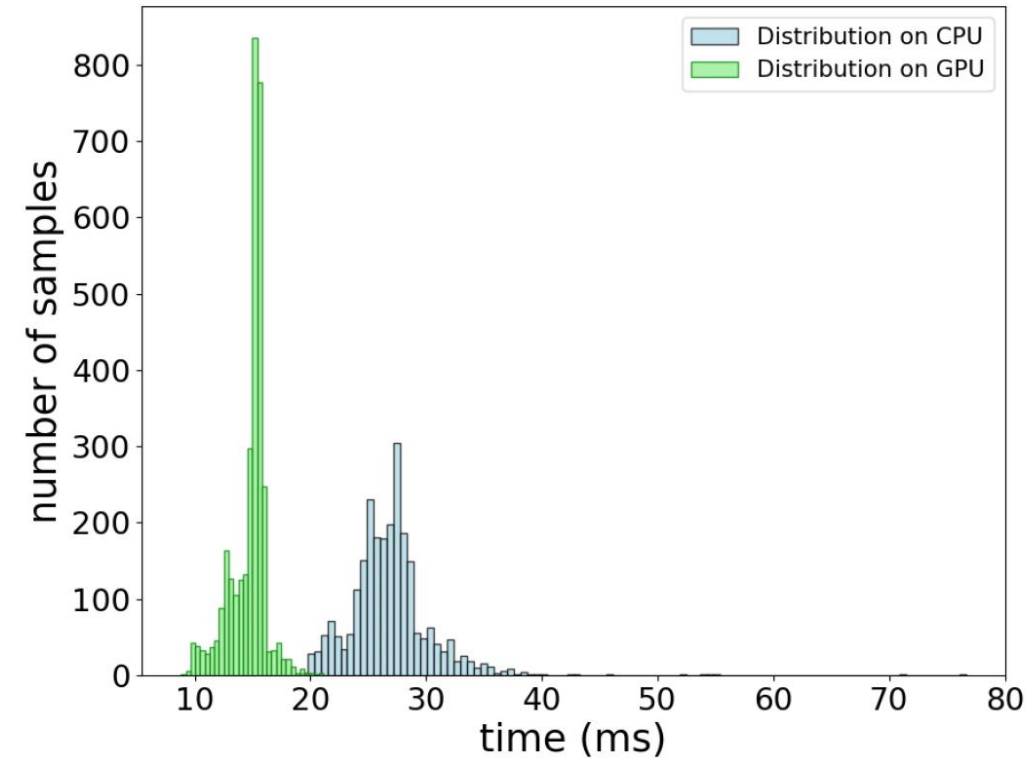
Transferability Across Beds

Table: Transferability Across Beds

Beds	Class	Metric	Fold1	Fold2	Fold3	Fold4	Fold5	Ave
H2F	ALL	Acc	93.90	92.84	88.90	90.98	94.74	93.99
		F1	90.20	90.71	91.92	88.86	94.73	92.46
	BRS	Acc	96.51	87.62	77.59	90.91	97.00	94.06
	NBR	Acc	90.71	94.52	99.25	92.20	96.44	94.41
F2H	SNR	Acc	94.49	96.39	89.86	89.82	90.78	93.50
	ALL	Acc	91.96	93.00	87.94	89.36	92.02	90.86
		F1	87.98	89.43	89.59	83.64	87.99	87.73
	BRS	Acc	93.02	91.75	80.67	87.23	84.13	87.36
	NBR	Acc	89.71	92.82	97.05	91.55	95.81	93.39
	SNR	Acc	93.14	94.42	86.09	89.30	96.13	91.82

Strong cross-bed generalization under patient-independent evaluation, with performance comparable to non-transfer settings.

Computational Efficiency and Real-Time Capability



GPU Speed: **14.78 ms** per sample.

CPU Speed: **26.95 ms** per sample.

Runs easily on low-power devices like Raspberry Pi

Fig. 7. Inference time distribution on two live experiments including both running on CPU and GPU.

Discussion and Future Work

Broader environmental validation:

Expand data collection to diverse bed and mattress types, in-home environments, and different sensor placements to further assess robustness under real-world conditions.

Dual-occupancy scenarios:

Explore multi-sensor configurations (e.g., both sides of the bed and bed frame) to mitigate signal overlap and improve individual signal separation in dual-occupancy settings.

Real-world deployment potential:

The robustness of seismic features for snoring and breath-holding supports continuous home deployment, with ongoing overnight studies validating long-term reliability.

Ablation on Data Preprocessing

Table: Ablation Study on Data Preprocessing

Method	Metric	Fold1	Fold2	Fold3	Fold4	Fold5	Ave
INT + DIFF	Acc	88.37	86.72	<u>84.60</u>	83.08	<u>84.74</u>	85.50
	F1	87.58	87.51	86.69	85.21	84.76	86.35
INT	Acc	82.79	85.97	80.64	<u>77.94</u>	84.76	82.42
	F1	<u>83.51</u>	86.47	84.87	<u>81.18</u>	77.63	<u>82.73</u>
DIFF	Acc	<u>85.67</u>	<u>86.43</u>	85.71	76.07	83.81	<u>83.54</u>
	F1	<u>83.40</u>	<u>87.43</u>	<u>86.62</u>	68.60	<u>83.30</u>	<u>81.87</u>
RAW	Acc	82.25	81.30	83.20	77.33	78.98	80.61
	F1	83.45	79.51	83.63	71.31	74.16	78.41

Raw performs worst. Combination of integral(int) and differential(diff) capture both low- and high-frequency information features provides a more comprehensive representation of the data. performs best.

Ablation on Model Structure

Table: Ablation Study on AFFCL

Model	Metric	Fold1	Fold2	Fold3	Fold4	Fold5	Ave
VANILLA	Acc	88.37	86.72	84.60	83.08	84.74	85.50
	F1	87.58	87.51	86.69	85.21	84.76	86.35
VANILLA + FAFS	Acc	90.09	87.12	85.10	85.12	87.31	86.95
	F1	<u>89.78</u>	88.34	86.90	86.04	88.58	87.93
VANILLA + ATTN	Acc	91.33	88.81	85.25	87.40	85.00	87.56
	F1	88.67	89.28	86.85	85.29	86.34	87.29
VANILLA + CL	Acc	90.25	86.95	85.83	87.86	87.21	87.62
	F1	88.28	88.62	86.59	87.64	88.27	87.88
VANILLA + FAFS + CL	Acc	<u>92.43</u>	90.62	88.27	91.28	93.53	91.23
	F1	<u>89.28</u>	88.95	90.09	89.50	90.10	89.58
VANILLA + ATTN + CL	Acc	91.64	93.95	88.80	89.04	92.70	91.23
	F1	87.96	89.61	91.46	<u>90.86</u>	<u>91.79</u>	<u>90.34</u>
VANILLA + FAFS + ATTN	Acc	91.82	88.58	<u>90.31</u>	<u>92.68</u>	<u>93.61</u>	<u>91.40</u>
	F1	89.49	<u>90.37</u>	<u>91.97</u>	87.86	90.86	90.11
VANILLA + FAFS + ATTN + CL	Acc	93.85	<u>93.12</u>	91.04	93.26	95.72	93.40
	F1	90.70	93.20	93.49	91.64	93.09	92.42

Compared with the vanilla model, adding any single submodule improves both accuracy and F1, while combining any two submodules yields further gains.

Incorporating all submodules achieves the best performance, outperforming the vanilla model by **+7.90% accuracy** and **+6.07% F1**.